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Part I

Pushover analyses

1 crushing compression — solved tension tail — perfect bond

1.1 Pushover — crushing/solved/perfect — mesh 25 mm — energy balance, uniaxial field — to 0.01% drift

2026-09-20_173801_pushover_crushing-solved-perfect_d0.01pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	25 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	7,446 / 32,423
composition	7,446 nodes, 32,423 elements (29,195 concrete, 1,740 longitudinal, 1,488 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 45.3, diagonal 32.4
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,560.6 \text{ mm}^2$
$E_t A_t$ per strut	48.4e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8$ MPa, $f_t = 2.03$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0411871, lambda:0.1, ft:2.03, Ets:521.981	394 concrete elements, L = 19 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0299316, lambda:0.1, ft:2.03, Ets:738.192	8 concrete elements, L = 26.87 mm
3	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0313916, lambda:0.1, ft:2.03, Ets:700.553	1,752 concrete elements, L = 25.5 mm
4	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0257189, lambda:0.1, ft:2.03, Ets:873.635	48 concrete elements, L = 31.8 mm
5	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0316751, lambda:0.1, ft:2.03, Ets:693.685	1,752 concrete elements, L = 25.25 mm
6	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258643, lambda:0.1, ft:2.03, Ets:868.137	48 concrete elements, L = 31.6 mm
7	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0333404, lambda:0.1, ft:2.03, Ets:655.91	3,504 concrete elements, L = 23.88 mm
8	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266878, lambda:0.1, ft:2.03, Ets:838.261	96 concrete elements, L = 30.51 mm
9	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0315799, lambda:0.1, ft:2.03, Ets:695.974	7,191 concrete elements, L = 25.33 mm
10	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258157, lambda:0.1, ft:2.03, Ets:869.968	564 concrete elements, L = 31.67 mm
11	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0319642, lambda:0.1, ft:2.03, Ets:686.817	102 concrete elements, L = 25 mm
12	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.026011, lambda:0.1, ft:2.03, Ets:862.659	8 concrete elements, L = 31.4 mm
13	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0232046, lambda:0.1, ft:2.03, Ets:981.067	48 concrete elements, L = 35.71 mm
14	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0233071, lambda:0.1, ft:2.03, Ets:976.174	48 concrete elements, L = 35.53 mm

tag	type	arguments	applied to
15	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0238799, lambda:0.1, ft:2.03, Ets:949.703	96 concrete elements, L = 34.57 mm
16	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0230714, lambda:0.1, ft:2.03, Ets:987.499	3,384 concrete elements, L = 35.94 mm
17	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0231719, lambda:0.1, ft:2.03, Ets:982.639	3,384 concrete elements, L = 35.77 mm
18	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0237331, lambda:0.1, ft:2.03, Ets:956.346	6,768 concrete elements, L = 34.81 mm
19	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	3,228 longitudinal/stirrup elements

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	8	corotTruss	19	1,186.1	31,030	3.680e7	1.937e6
concrete	4	corotTruss	19	1,388.9	31,030	4.310e7	2.268e6
concrete	20	corotTruss	19	1,591.8	31,030	4.939e7	2.600e6
concrete	4	corotTruss	19	1,584	31,030	4.915e7	2.587e6
concrete	20	corotTruss	19	1,576.2	31,030	4.891e7	2.574e6
concrete	4	corotTruss	19	1,533.3	31,030	4.758e7	2.504e6
concrete	46	corotTruss	19	1,490.4	31,030	4.625e7	2.434e6
concrete	2	corotTruss	19	1,383.7	31,030	4.294e7	2.260e6
concrete	188	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	92	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	2	corotTruss	19	1,571	31,030	4.875e7	2.566e6
concrete	4	corotTruss	19	1,560.6	31,030	4.843e7	2.549e6
concrete	48	corotTruss	23.88	1,186.1	31,030	3.680e7	1.541e6
concrete	48	corotTruss	23.88	1,560.6	31,030	4.843e7	2.028e6
concrete	24	corotTruss	23.88	1,383.7	31,030	4.294e7	1.798e6
concrete	2,256	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	1,104	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	24	corotTruss	23.88	1,571	31,030	4.875e7	2.041e6
concrete, orthogonal	4	corotTruss	25	1,186.1	31,030	3.680e7	1.472e6
concrete, orthogonal	4	corotTruss	25	1,388.9	31,030	4.310e7	1.724e6
concrete, orthogonal	20	corotTruss	25	1,591.8	31,030	4.939e7	1.976e6
concrete, orthogonal	4	corotTruss	25	1,584	31,030	4.915e7	1.966e6
concrete, orthogonal	20	corotTruss	25	1,576.2	31,030	4.891e7	1.956e6
concrete, orthogonal	4	corotTruss	25	1,533.3	31,030	4.758e7	1.903e6
concrete, orthogonal	46	corotTruss	25	1,490.4	31,030	4.625e7	1.850e6
concrete	24	corotTruss	25.25	1,186.1	31,030	3.680e7	1.458e6
concrete	24	corotTruss	25.25	1,560.6	31,030	4.843e7	1.918e6
concrete	12	corotTruss	25.25	1,383.7	31,030	4.294e7	1.700e6
concrete	1,128	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	552	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	12	corotTruss	25.25	1,571	31,030	4.875e7	1.931e6
concrete	282	corotTruss	25.33	1,186.1	31,030	3.680e7	1.453e6

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	282	corotTruss	25.33	1,388.9	31,030	4.310e7	1.701e6
concrete	1,410	corotTruss	25.33	1,591.8	31,030	4.939e7	1.950e6
concrete	282	corotTruss	25.33	1,584	31,030	4.915e7	1.940e6
concrete	1,410	corotTruss	25.33	1,576.2	31,030	4.891e7	1.931e6
concrete	282	corotTruss	25.33	1,533.3	31,030	4.758e7	1.878e6
concrete	3,243	corotTruss	25.33	1,490.4	31,030	4.625e7	1.826e6
concrete	24	corotTruss	25.5	1,186.1	31,030	3.680e7	1.443e6
concrete	24	corotTruss	25.5	1,560.6	31,030	4.843e7	1.899e6
concrete	12	corotTruss	25.5	1,383.7	31,030	4.294e7	1.684e6
concrete	1,128	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	552	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	12	corotTruss	25.5	1,571	31,030	4.875e7	1.912e6
concrete	8	corotTruss	26.87	1,186.1	31,030	3.680e7	1.370e6
concrete	96	corotTruss	30.51	1,329.5	31,030	4.126e7	1.352e6
concrete	8	corotTruss	31.4	1,360.5	31,030	4.222e7	1.344e6
concrete	48	corotTruss	31.6	1,367.3	31,030	4.243e7	1.343e6
concrete	564	corotTruss	31.67	1,369.5	31,030	4.250e7	1.342e6
concrete	48	corotTruss	31.8	1,374	31,030	4.264e7	1.341e6
concrete	96	corotTruss	34.57	1,525.1	31,030	4.732e7	1.369e6
concrete	6,768	corotTruss	34.81	1,535.2	31,030	4.764e7	1.369e6
concrete, diagonal	48	corotTruss	35.53	1,568.4	31,030	4.867e7	1.370e6
concrete	48	corotTruss	35.71	1,576.1	31,030	4.891e7	1.370e6
concrete	3,384	corotTruss	35.77	1,578.8	31,030	4.899e7	1.370e6
concrete	2,256	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
concrete	1,128	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
longitudinal	16	corotTruss	19	142.7	200,000	2.853e7	1.502e6
longitudinal	8	corotTruss	19	63.3	200,000	1.267e7	6.667e5
longitudinal	16	corotTruss	25	142.7	200,000	2.853e7	1.141e6
longitudinal	8	corotTruss	25	63.3	200,000	1.267e7	5.067e5
longitudinal	1,128	corotTruss	25.33	142.7	200,000	2.853e7	1.126e6
longitudinal	564	corotTruss	25.33	63.3	200,000	1.267e7	5.001e5
stirrup	456	corotTruss	23.88	63.3	200,000	1.267e7	5.305e5
stirrup	228	corotTruss	25.25	63.3	200,000	1.267e7	5.017e5
stirrup	576	corotTruss	25.5	35.6	200,000	7.118e6	2.791e5
stirrup	228	corotTruss	25.5	63.3	200,000	1.267e7	4.968e5

Analysis and solver

analysis	pushover
target	0.01% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	2.739 us, 7,305 steps per fundamental period
first period T_1	20.009 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	17,583
wall clock	0.1 h
converged	True

Recorded response

peak base shear	9.8 kN (5 ms moving average)
raw sample maximum	10.4 kN
drift at peak	0.009%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.01% drift
inertia + damping in the reaction,	2.2 kN
ascending branch (p95)	

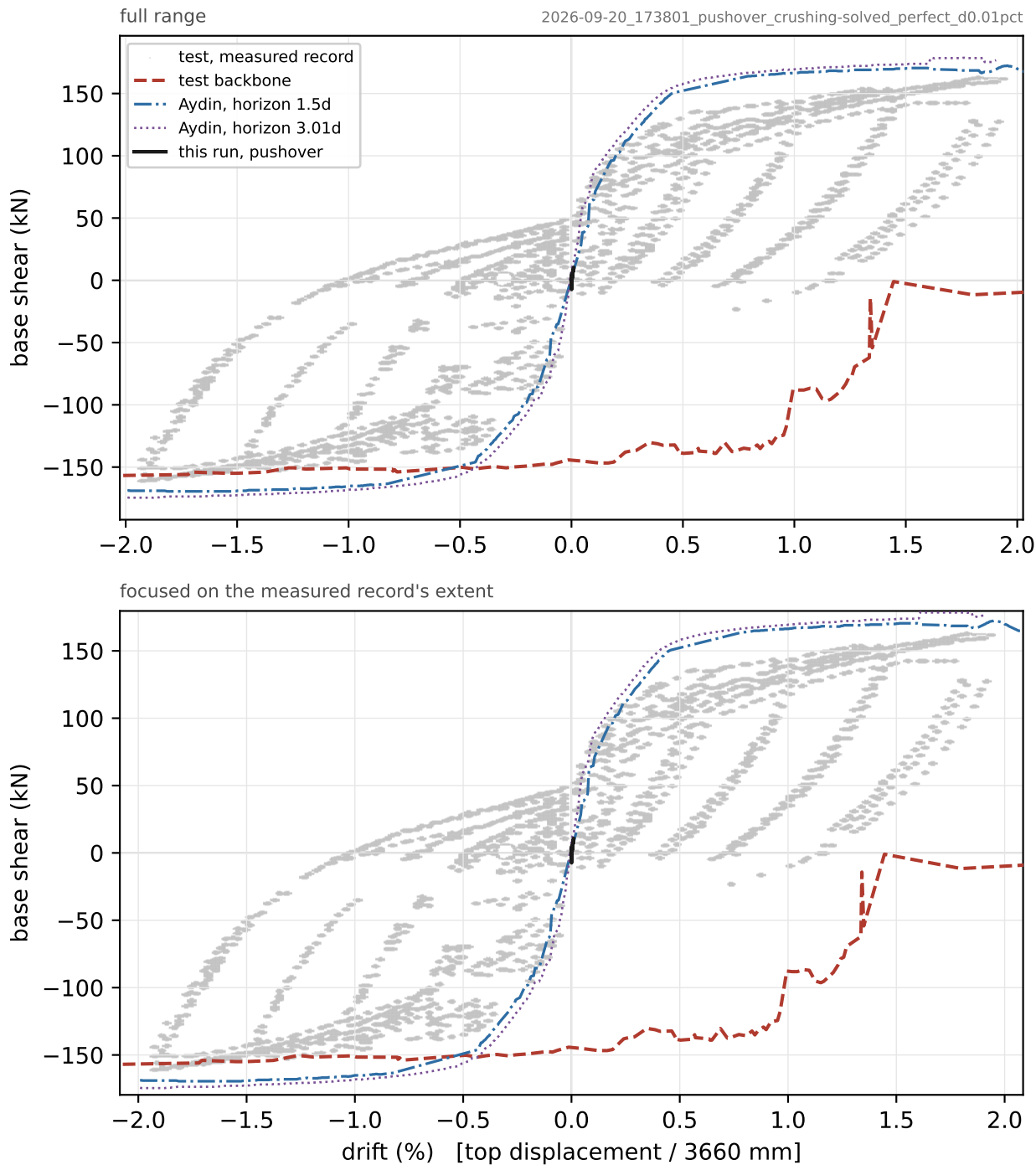


Figure 1: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

1.2 Pushover — crushing/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — to 2.5% drift

2026-09-20_174437_pushover_crushing-solved_perfect_guniform_m30.5_d2.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,002
composition	4,961 nodes, 22,002 elements (19,360 concrete, 1,440 longitudinal, 1,202 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266976, lambda:0.1, ft:2.03, Ets:837.916	9,760 concrete elements, L = 30.5 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.022069, lambda:0.1, ft:2.03, Ets:1184.99	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,642 longitudinal/stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	9,760	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	9,600	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	722	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	pushover
target	2.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	4.416 us, 4,556 steps per fundamental period
first period T_1	20.12 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	2,726,278
wall clock	13.32 h
converged	True

Recorded response

peak base shear	139.4 kN (5 ms moving average)
raw sample maximum	162.3 kN
drift at peak	0.497%

drift capacity (20% drop) 2.374% — basis: monotonic
 traced to 2.5% drift
 inertia + damping in the reaction, 1.3 kN
 ascending branch (p95)

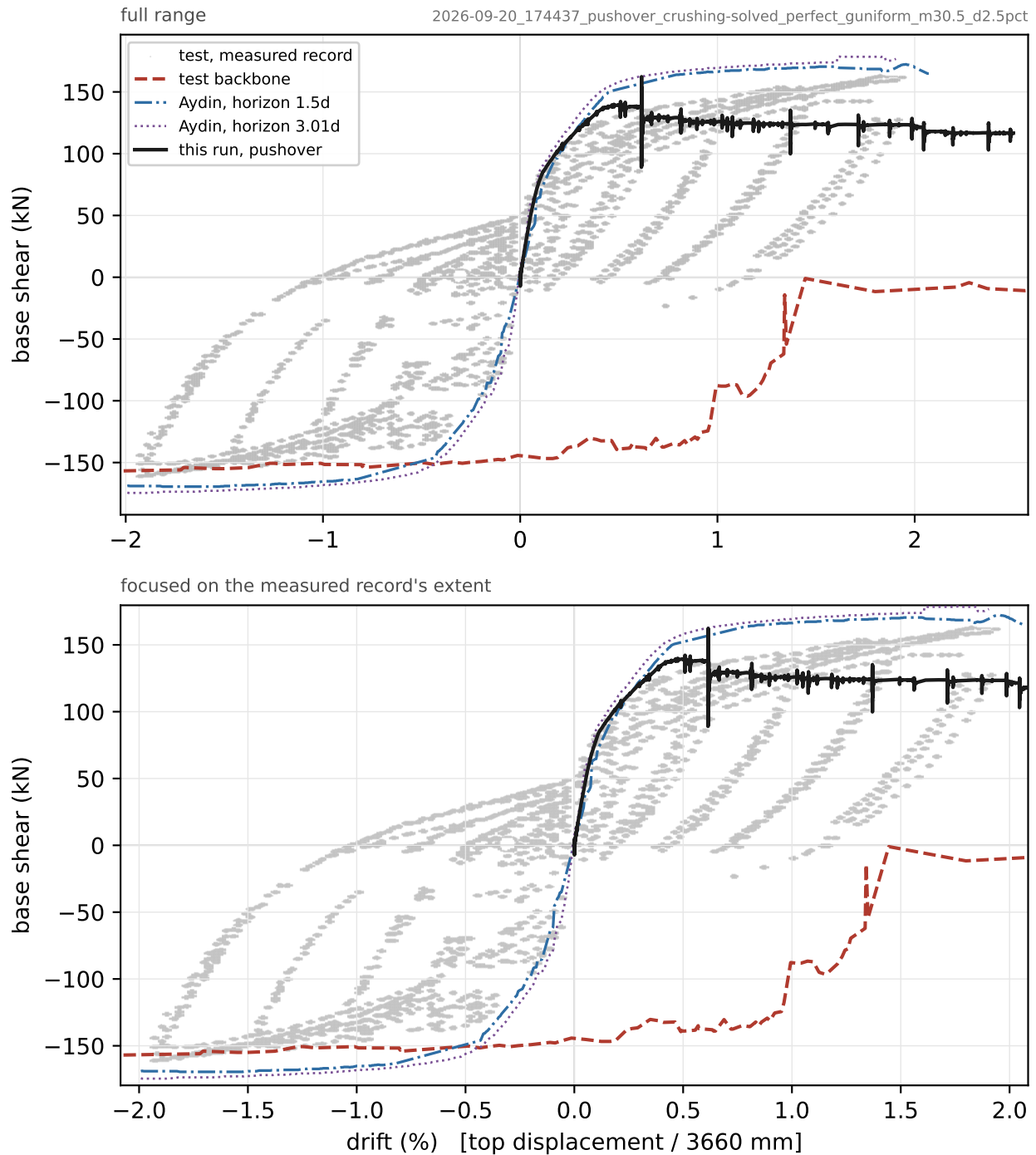


Figure 2: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.